Economics — Adaptive Study App

User Guide · Grade 9 IGCSE Economics · Personal local study tool

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1. What this app does

This is a personal, offline study tool built around your IGCSE Economics syllabus. Instead of working through the textbook chapter by chapter in order, it works the way apps like Math Academy do: it tracks what you've actually mastered, only unlocks a new topic once you've proven you know its prerequisites, and brings back older topics for review on a schedule so you don't forget them.

Everything runs in your web browser, on your own computer. No account, no internet connection required for day-to-day studying, and no data is sent anywhere — your progress is stored only in that browser.

2. Starting the app

Everyday use — just double-click

Open File Explorer, go to `D:\economics\app`, and double-click `index.html`. It opens in your default browser and you're ready to go. Do this every time you want to study — it's the normal way to use the app.

Optional: running it via a local server

You only need this if you want to use the microphone (voice input) feature — otherwise, skip this section entirely.

> Chrome and Edge block microphone access on pages opened directly from a file, so voice input needs the app served over `localhost` instead.

Open Command Prompt and run:

```
cd /d D:\economics\app
python -m http.server 8791
```

Then open `http://localhost:8791` in Chrome or Edge. Leave that Command Prompt window open while you study — closing it stops the server. Voice input also needs an active internet connection, since the speech recognition itself runs in the cloud, not on your computer.

> **Common mistake:** typing `cd D:\economics\app` alone doesn't switch drives if you're currently on drive C: — it silently stays put. Always use `cd /d D:\economics\app` (with the `/d` flag) instead.

3. The dashboard, explained

When you open the app, you land on the dashboard — a list of all 36 syllabus topics grouped by section, plus a button to start today's session.

What you see	What it means
locked	You haven't unlocked this topic yet — one or more prerequisite topics still need to be mastered first.
available	Unlocked and ready to practice. You'll see a mastery percentage bar next to it.

mastered	You've proven mastery across two separate sessions. Shows a "next review in N days" note instead of a live mastery bar.
review due	A mastered topic has come back around on its spaced-review schedule and is ready to be tested again.
"N to retest" (red)	You're below 80% mastery on this topic and have specific questions you previously got wrong, queued up to try again.
"no content yet" (amber)	This topic exists in the syllabus map but doesn't have questions written for it yet.

The **"Start today's session"** button tells you up front how many items are queued, split into due reviews vs. in-progress topics, e.g. *"3 items — 1 due for review, 2 in progress."* The button is greyed out if there's genuinely nothing to do right now.

4. Running a study session

Click "Start today's session" and you'll be taken through one topic at a time, one question at a time. Each question shows a difficulty tag (easy / med / hard) and is one of two types:

- **Multiple choice:** click an option — it's graded instantly, with the correct answer highlighted and an explanation shown.
- **Short answer:** type (or speak — see section 10) your answer, then click **"Reveal answer"** to see the model answer and explanation. Honestly self-mark it as **"I got it right"** or **"I got it wrong"** — the app relies on your own honesty here, the same way flashcard apps do.

At the end of the queue, you'll see a summary: how many you got right out of the total, which topics you touched, and your streak.

5. How topics unlock (mastery gating)

A topic only becomes "available" once every one of its listed prerequisites is "mastered." For example, Demand and Supply both require Opportunity Cost and Markets to be mastered first, and Price Elasticity of Demand requires both Demand and Price Determination.

To go from "available" to "mastered," you need to score **80% or higher across two separate sessions** on that topic — not just one lucky session. If a session dips below 80%, your progress toward mastery resets, but your questions-in-progress and mastery percentage are remembered, so nothing is lost — you just need another good session before it "counts."

6. Spaced review

Once a topic is mastered, it doesn't disappear — it comes back for review on an increasing schedule: 1 day after mastering, then 3, then 7, then 16, then 35 days, as long as you keep answering correctly. Get a review question wrong, and the schedule resets back to 1 day, and if your mastery on that topic drops far enough, it can even fall out of "mastered" status and go back into active practice.

This is the same idea behind spaced-repetition flashcard apps: the goal is to test you right before you'd naturally start forgetting something, not on a fixed daily basis regardless of how well you know it.

7. Being retested on what you got wrong

If you're below 80% mastery on a topic, the app doesn't just throw random questions at you again — it specifically remembers which questions you got wrong and puts those first in your next session on that topic. Once the whole question bank for a topic has been shown at least once, it cycles back through from the start rather than repeating the same handful over and over.

8. Your streak

The 🔥 badge in the top-right of the dashboard shows how many days in a row you've completed at least one session. It only counts once per day (extra sessions the same day don't inflate it), and if you miss a day, it resets — though your all-time best streak is always remembered and shown when you hover over the badge.

9. The Economics Dictionary

Click **"📖 Dictionary"** at the top of the dashboard to open an alphabetical glossary of key economics terms covered so far (currently over 60 terms). Use the search box at the top to filter by term or definition — useful for quickly looking something up mid-revision without digging through the textbook.

10. Voice input for typed answers

On short-answer questions, you'll see a 🎤 button next to the text box. Click it, speak your answer, and it gets transcribed into the box for you.

> This only works when the app is opened via `http://localhost:8791` (see section 2) using Chrome or Edge, **with an active internet connection**. If you open the app the normal way (double-clicking `index.html`), the button will appear greyed out with an explanation — typing still works exactly the same either way.

11. Where your progress is saved

All your mastery levels, streak, and history are stored in your browser's local storage — nothing is uploaded anywhere. A few practical consequences:

- Progress is tied to the specific **browser** you use. If you switch from Chrome to Edge, or use a different computer, you start fresh in that browser.
- Opening the app via `index.html` directly and via `localhost:8791` count as *different* storage locations — so progress made one way won't show up the other way. Pick one method as your main way of studying to avoid confusion (double-clicking is fine for daily use).
- Clearing your browser's site data/history for this app would erase your progress, so avoid doing that if you want to keep it.

12. Troubleshooting

Problem	Fix
Browser shows a plain file listing instead of the app	You're in the wrong folder in Command Prompt. Use `cd /d D:\economics\app` (with `/d`) before starting the server.
Mic button is greyed out	You're either using `file://` directly, or a browser other than Chrome/Edge. Start the local server and use Chrome or Edge instead.
Clicking the mic does nothing / no transcription appears	Check you allowed microphone permission when prompted, and that you have an internet connection — the speech recognition itself is cloud-based.
A topic says "no content yet"	Questions haven't been written for that topic yet — ask for it to be added.
Progress seems to have disappeared	You're likely opening the app a different way than before (file:// vs localhost), or in a different browser. See section 11.

13. What's covered so far

Question banks currently exist for topics 1 through 11 (The Basic Economic Problem through Price Elasticity of Supply), with extra depth on Demand, Supply, PED and PES. Topics 12 onward exist in the syllabus map (so the unlock structure is ready) but don't have questions yet. Just ask for the next topic whenever you're ready to keep going.